The Winning Solution for Track 1 of the 2026 AI City Challenge: Gauge Canonicalization for Multi-Camera 3D Perception

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Abstract. Pretrained multi-camera 3D models are often deployed to new sites without labels or retraining. Although sim-to-real work usually emphasizes appearance, we identify a geometric failure in an anchor-seeded Sparse4D detector: its learned spatial anchors are tied to the training coordinate frame, so moving the world origin can suppress all detections. We formalize this anchor–gauge coupling as a failure of equivariance under horizontal translations. GaugeAlign restores this equivariance through closed-form scene canonicalization at inference time. We evaluate protocol alignment and a temporal-prior adjustment as separate system components. Canonicalization uses known camera calibration, is exactly equivariant to horizontal translations, and requires neither target labels nor weight updates. Our submitted online system won Track 1 of the 2026 AI City Challenge, scoring 56.54 HOTA on the hidden test set, 4.53 points above the runner-up. Controlled validation isolates the geometric effect: reverting to the absolute target gauge lowers mean DetA from 36.16 to 12.25 and yields zero detections on one four-camera scene. This causal attribution is limited to the validation study; the hidden-test leaderboard evaluates the complete system. The source code is publicly available at https://github.com/HyperbolicCurve/GaugeAlign-AICity2026.git.

Keywords: Multi-camera 3D tracking · Sim-to-real transfer · Coordinate frames · Canonicalization · Query-based detection · Spatial priors

1 Introduction

Multi-camera 3D perception systems for warehouses and factories are rarely trained per site. A model is pretrained once, largely on synthetic data, and transferred to each new camera installation, ideally without labels or retraining. The obstacle to such transfer is usually framed as an appearance gap, and the standard remedies operate on images: domain randomization, generative augmentation, and real data in pretraining. Track 1 of the 2026 AI City Challenge [29] follows this premise. It asks for online RGB-only 3D detection and tracking of

seven object classes across 4–20 calibrated cameras, scored by 3D HOTA [19] on a hidden test set that mixes synthetic and real warehouses.

Our starting point is an appearance-adapted baseline. NVIDIA's TAO release provides Sparse4D warehouse models [23] from the query-based multi-view detector family [17]. These models were pretrained on the prior-year synthetic benchmark, a COSMOS-augmented version of that benchmark [22], and real data from several environments. The released checkpoint scores 50.99 HOTA on the 2025 benchmark. Yet naive transfer to one 2026 scene produces no boxes over an entire four-camera sequence, with a peak class confidence of about 0.03. This failure cannot be explained by appearance alone.

Sparse4D seeds its decoder with 900 anchor boxes obtained by k-means over training-scene object centers. This prior is expressed in the training world frame, whose scenes place the origin near the cameras. The different absolute origins at the 2026 sites displace this fixed prior relative to the calibrated camera frustums. In the four-camera case, enough sampling points fall outside the visible region that the decoder collapses. We call this failure anchor–gauge coupling: a learned spatial prior is tied to the world-coordinate convention used during training, while that convention need not be preserved across sites.

GaugeAlign addresses this failure at inference time. Its core uses the mean camera position, computed from calibration, to express a scene in a canonical frame; the exact inverse returns predictions to the official frame. A class-ID map aligns the challenge protocol, while the instance-bank decay is tuned separately as a temporal prior. The coordinate round trip is numerically exact. On controlled validation scenes, removing canonicalization cuts mean DetA by two thirds. On the hidden test set, the complete system ranks first among ten entries at 56.54 HOTA, 4.53 above the runner-up. Its DetA is 10.3 points higher, whereas its AssA is 7.1 points lower.

In summary, the main contributions are as follows: **1)** We identify and formalize anchor–gauge coupling, a geometric failure mode in cross-site transfer. **2)** We introduce GaugeAlign, a calibration-driven inference-time framework built on a closed-form canonicalization operator with an exact guarantee of equivariance to horizontal translations. **3)** We isolate the alignment on validation data and report the winning hidden-test system separately. **4)** We report negative fine-tuning and association results that clarify the framework's limits.

2 Related Work

2.1 Multi-camera 3D detection and tracking

DETR casts detection as set prediction over learned object queries [2, 39]. Its multi-camera 3D descendants differ mainly in where those queries live. DETR3D projects learned 3D reference points into the images [36], BEVFormer pools features onto a bird's-eye-view grid with a fixed metric range [13], PETR attaches 3D position encodings [18], and Sparse4D propagates sparse instance queries through time [16, 17]. Fixed BEV ranges, reference points, and position encodings may exhibit related frame sensitivity, but our experiments establish it only

for Sparse4D's k-means anchors. The outside-in setting goes back to multiview pedestrian detection with static camera networks [3,7] and is now driven by the AI City Challenge [29,30]. The strongest 2025 Track 1 systems fuse RGB-D point clouds: the winning V-DETR pipeline with point-cloud re-identification [12], the cluster-and-BEV runner-up [32], and the third-place 2D-to-3D late-aggregation system [11]. Their use of depth or offline processing places them outside the online RGB-only setting studied here. These systems use calibration to back-project depth or 2D detections, map them into a common world frame, and fuse or associate the resulting geometry [11,12,32]. GaugeAlign instead changes the coordinate representation presented to a frozen detector and exactly inverts that change at its output. Calibration is therefore common in challenge systems, but GaugeAlign uses it at a different stage of the pipeline. Closest to our work, a concurrent NVIDIA framework adapts Sparse4D to warehouse camera networks with absolute world-coordinate priors, a learned re-identification head, and COSMOS augmentation, reaching the best online camera-only result on the 2025 benchmark [35], with a companion study of its quantized, reduced-frame-rate deployment [1]. Such absolute-coordinate priors can break under cross-site transfer. GaugeAlign instead re-gauges the input so that the learned prior aligns with the target scene.

2.2 Label-free adaptation and transfer

When a symmetry is hard to build into the architecture [4], an alternative is to canonicalize the input: spatial transformers learn such warps [8], alignment networks do the same for point clouds [25], and later work formalizes learned canonicalization [9], frame averaging [24], and adapters that equip frozen pre-trained models with a symmetry [20]. All of these train the canonicalizer. Ours is closed-form and computed from calibration; it restores horizontal-translation equivariance broken by a learned anchor prior. Test-time adaptation shares the no-label constraint but updates model state on target data, through entropy minimization [34], normalization statistics [27], self-supervision [28], or source-free transfer [15]. Liang et al. [14] survey this line. Because gauge canonicalization has no learned state and is exactly invertible, it avoids the parameter drift associated with test-time adaptation. On the appearance side, sim-to-real transfer builds on domain randomization [31,33], rendered corpora [26,30], and generative augmentation with world models [21,22]. Generative models also hold implicit 3D priors for scene understanding [37], and the same generalization pressure appears in robotic manipulation in dynamic environments [5]. The baseline was already pretrained with appearance-focused transfer strategies, allowing us to isolate the residual geometric term. Fine-tuning can cause catastrophic forgetting [10], and our experiments show severe degradation at practical batch sizes. HOTA's DetA/AssA decomposition separates the detection improvement from the remaining association error [19].

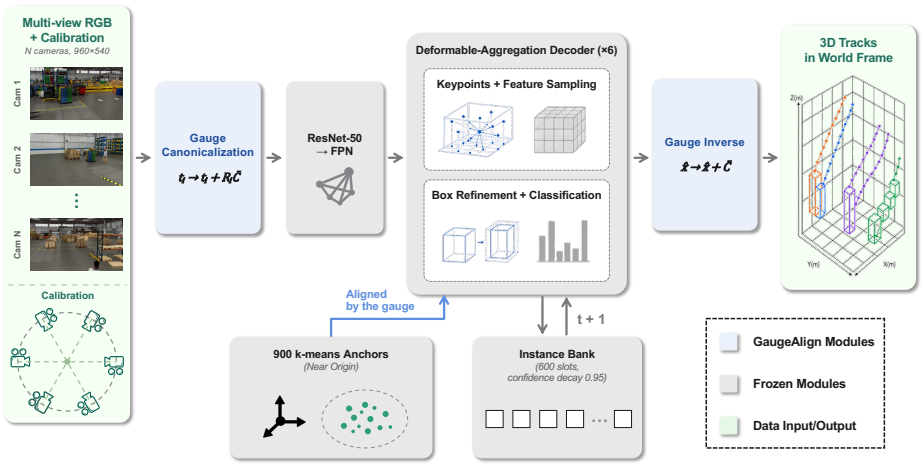


Fig. 1: GaugeAlign overview. "Frozen modules" (gray)—backbone/FPN, decoder, 900 k-means anchors, and instance bank—keep the public checkpoint's architecture and learned parameters. The bank's non-learned decay is configured separately. Blue modules change only calibration and predicted centers: $t_i \to t_i + R_i \bar{C}_{xy}$ before inference, followed by the exact inverse. The arrow marks anchor–training-frame coupling; the schematic's $\bar{C}$ denotes $\bar{C}_{xy}$.

3 The GaugeAlign Framework

GaugeAlign aligns three interfaces between a pretrained multi-camera detector and a target site (Fig. 1): calibration defines the coordinate gauge (Sec. 3.3), a fixed map aligns the class taxonomy (Sec. 3.4), and the instance-bank decay controls the tracker's temporal prior (Sec. 3.5).

3.1 Problem setting

A scene provides N static cameras with known intrinsics and extrinsics (R_i, t_i). The task is online 3D tracking. Per frame, the system outputs 3D boxes in a shared world frame, each parameterized as $(x, y, z, w, l, h, \text{yaw})$ and carrying a class ID and a track ID consistent across all cameras. Inference is RGB-only, and we target the online setting of the track, which forbids the use of future frames. Evaluation uses 3D HOTA with 3D-IoU matching, and its decomposition into DetA (detection), AssA (association) and LocA (localization) drives our analysis.

The target detector is a Sparse4D warehouse model [23] with a ResNet-50 backbone [6]. Four properties matter. First, the input is a single synchronized tensor over the N cameras ($N{\times}3{\times}540{\times}960$) with per-camera projection matrices, so all geometry enters through the extrinsics. Second, the decoder is seeded by 900 anchors obtained by k-means over training-scene object centers, a spatial prior expressed in the scene-centered training frame. Third, an instance bank caches up to 600 temporal instances and decays a per-instance confidence while an object goes unmatched, so one constant governs identity survival. Fourth,

the pretraining mix (a synthetic warehouse benchmark, a COSMOS-augmented copy, and real multi-environment data) targets appearance transfer explicitly. We use this appearance-adapted checkpoint to isolate residual geometric failures.

3.2 The anchor–gauge coupling

Write a scene as $S = (\{I_i\}, \{K_i, R_i, t_i\})$: synchronized images I_i from cameras that map a world point $x \in \mathbb{R}^3$ to camera coordinates $R_i x + t_i$ and, through the intrinsics K_i and the perspective projection π, to a pixel. Call the choice of world origin the *gauge* of the scene. The submitted operator acts on horizontal gauge translations $d = (d_x, d_y, 0)$. Such a translation acts on the scene and on any set of boxes B as

$$\tau_d S = \big(\{I_i\}, \{K_i, R_i, t_i - R_i d\}\big), \qquad \tau_d B = \{\, b + d \,\}, \tag{1}$$

where $b + d$ shifts a box center and leaves size and yaw unchanged. The images are untouched, ground truth and calibration transform together, and 3D-IoU is preserved, so the task itself is gauge invariant. An ideal detector f, mapping a scene to a set of world-frame boxes, would therefore be equivariant under horizontal translations,

$$f(\tau_d S) = \tau_d\, f(S) \quad \text{for all } d = (d_x, d_y, 0). \tag{2}$$

An anchor-seeded query detector violates Eq. (2) structurally. Its decoder starts from a fixed anchor set $A = \{a_k\}$, whose box centers x_k are stored in the coordinates of the training gauge. At every layer, keypoints derived from the current box estimate are projected into each view,

$$u_{ik} = \pi\big(K_i(R_i x_k + t_i)\big), \tag{3}$$

image features are sampled at the u_{ik}, and the box is refined by a local update. Under a gauge translation the projection argument becomes $R_i x_k + (t_i - R_i d) = R_i(x_k - d) + t_i$, so shifting the frame shifts every anchor by $-d$ relative to the scene. Because A is a network constant, Eq. (2) fails. As overlap between projected anchors and camera frustums decreases, more samples return background, and local refinement cannot reliably recover a scene-wide displacement. The offset at which this becomes a complete failure depends on scene and camera layout.

Naive transfer produces exactly this condition. Let $C_i = -R_i^\top t_i$ be the camera centers and $\bar{C}_{xy} = N^{-1} \sum_i (C_{ix}, C_{iy}, 0)$ their horizontal centroid. In the released 900-anchor file, center coordinates span $[-80.2, 86.4]$ m in x and $[-125.6, 124.7]$ m in y, but their median is $(2.3, -0.4)$ m, consistent with a scene-centered training convention. The target validation scenes have $|\bar{C}_{xy}| \approx 86$–$104$ m. The issue is therefore not that every anchor lies outside a numerical range; it is the displacement of the anchor distribution relative to each scene's calibrated visible region. Figure 2 shows both gauges on the same frame with the same weights: the canonical gauge recovers objects, whereas the absolute frame yields no detections over the whole sequence in W022.

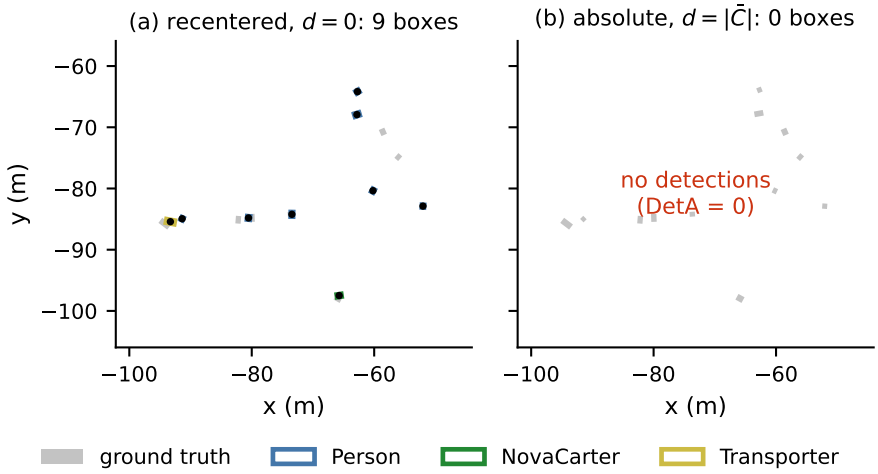


Fig. 2: The anchor–gauge coupling on a four-camera validation scene (Warehouse 022, busiest frame, bird's-eye view). Gray: ground truth; colored: predictions. (a) The canonical gauge recovers the objects. (b) In the absolute frame, the same detector emits zero boxes from the same images throughout the sequence. Only the coordinate frame differs. In this original graphic, scalar d denotes δ, and $|\bar{C}|$ denotes $|\bar{C}_{xy}|$.

3.3 The scene canonicalization operator

Horizontal equivariance can be restored by canonicalizing the input of f. The horizontal camera centroid is an equivariant functional of the scene, $\bar{C}_{xy}(\tau_d S) = \bar{C}_{xy}(S) + d$, so it provides a reference point that moves with the gauge. We define the canonicalized detector

$$F(S) \;=\; \tau_{\bar{C}_{xy}(S)}\, f\big(\tau_{-\bar{C}_{xy}(S)}\, S\big), \tag{4}$$

which runs the detector in the camera-centroid frame and shifts the predictions back. F satisfies Eq. (2) exactly. Using $\tau_a \tau_b = \tau_{a+b}$ and the equivariance of $\bar{C}_{xy}$,

$$F(\tau_d S) = \tau_{\bar{C}_{xy}(S)+d}\, f\big(\tau_{-\bar{C}_{xy}(S)-d}\, \tau_d S\big) = \tau_d\, \tau_{\bar{C}_{xy}(S)}\, f\big(\tau_{-\bar{C}_{xy}(S)}\, S\big) = \tau_d\, F(S). \tag{5}$$

The construction parallels canonicalization methods that fix a symmetry for a non-equivariant backbone, but the frame functional is closed-form rather than learned. By Eq. (1), the canonical scene replaces each extrinsic translation by

$$t_i' = t_i + R_i\, \bar{C}_{xy}, \tag{6}$$

and the inverse adds $\bar{C}_{xy}$ back to every predicted box center.

The operator has three properties. It is *label-free*: $\bar{C}_{xy}$ uses no annotations. It is *exact*: the round trip changes box coordinates by less than 10^{-6} m, so the transform itself does not alter scoring. It is *plug-in*: it acts only on calibration and predicted centers. In the controlled ablation, where all other settings are fixed, the score difference is therefore attributable to the gauge change.

3.4 Design analysis

Choice of the canonical center. The ideal center would reproduce the unknown training convention, and the mean horizontal camera position is one label-free estimate. Along the single tested direction, performance changes by less than one DetA point from 0 to 25 m. This shows tolerance around the chosen mean, but does not establish that a median or frustum-intersection center performs equally; those alternatives remain untested.

Scope of the failure mode. Our diagnosis is demonstrated for one anchor-seeded Sparse4D model. Fixed BEV ranges, learned reference points, or position embeddings could exhibit a similar dependence if expressed in a site-specific world frame, but this is a hypothesis rather than an empirical result. Concurrent outside-in work also uses absolute-coordinate priors [35]. The present operator handles horizontal translation only; vertical shifts, rotations, tilt bias, axis conventions, and scale are outside its guarantee.

Calibration requirement. GaugeAlign assumes target-site intrinsics and extrinsics are available and sufficiently accurate. Calibration is already an input to the challenge detector, but GaugeAlign cannot be applied as specified to an uncalibrated rig. Sensitivity to calibration error is not evaluated and remains a practical limitation.

Protocol alignment layer. The detector's class indices differ from the official Track 1 ids and are remapped (0→0, 1→4, 2→5, 3→2, 4→3, 5→1, 6→6). Removing this correction costs 2.83 DetA on the four-camera validation scene, whereas removing canonicalization causes a complete collapse.

3.5 Temporal prior adaptation

The anchors are not the only prior tuned to the training distribution. While an instance goes unmatched, the bank multiplies its confidence by a decay constant λ per frame, $c^{(t+1)} = \lambda\, c^{(t)}$, and drops the identity once the confidence falls below the emission threshold. The survival time under occlusion scales as $1/\ln(1/\lambda)$, so λ is a temporal prior on occlusion length. In our submission, raising λ from 0.8 to 0.95 increases the nominal occlusion horizon by $\ln 0.8/\ln 0.95 \approx 4.4\times$ and improves the hidden-test score by 0.54 HOTA. This effect is small relative to the gauge effect. We compare $\lambda \in \{0.9, 0.95\}$ against the native value of 0.8 and use 0.95 in the final submission.

Inference procedure. For each scene, compute $\bar{C}_{xy} = N^{-1}\sum_i (C_{ix}, C_{iy}, 0)$ once, replace t_i by $t_i + R_i\bar{C}_{xy}$, and set the chosen instance-bank decay. Frames are then processed in temporal order. For every output box b, add $\bar{C}_{xy}$ to its center and apply the class-ID map σ before emission. The added work is one $O(N)$ transform per scene and one vector addition per box.

4 Experiments

4.1 Setup

Benchmark. We evaluate on Track 1 of the 2026 AI City Challenge [29]: seven object classes (Person, Forklift, NovaCarter, Transporter, FourierGR1T2, AgilityDigit, PalletTruck), 4–20 cameras per scene, and a hidden test set of five scenes, three synthetic and two real. 3D HOTA is computed per class within each scene and weighted across scenes by object count. HOTA, DetA, AssA, and LocA are aggregated independently over thresholds, classes, and scenes, so HOTA is not the square root of the printed aggregate DetA and AssA. A 50% subset drove development, and the final ranking uses the full set. The tables reproduce server-displayed metrics without post-hoc scaling; the rules separately assign a 1.1 award factor to online entries. All ten final entries were marked online, so the factor does not change their order. Hidden-test scores mix synthetic and real scenes, so we make no real-only claim. Controlled experiments use Warehouse 020/021/022 with 16, 16, and 4 cameras.

Implementation. We instantiate GaugeAlign on the publicly released NVIDIA TAO Sparse4D checkpoint (ResNet-50, version `deployable_v2.2`) [23], with detector parameters fixed. Inference runs at 960×540 (identical to full resolution on our scenes), at native frame rate, with per-class score thresholds and a floor-region filter. Because post-processing may use future frames, online status is a property of the entire pipeline rather than the base detector alone. We verify the final online configuration with a prefix-invariance test: its output through a cutoff is identical whether inference is run on that prefix or on the full sequence.

Evaluation protocol. Hidden-test submissions are scarce, so we follow a fixed discipline: the GaugeAlign base score serves as a reference baseline, and every subsequent configuration changes one variable against a named control and must first show no DetA or LocA regression on the validation scenes. One early configuration with a flawed size prior, submitted before this check was in place, scored 53.78; no configuration that passed the check fell below the baseline.

4.2 Main results

Main result. Table 1 reports the submitted progression. The base configuration, canonicalization with protocol alignment, scores 56.00, and the temporal prior adaptation (λ=0.95) adds 0.54. The resulting 56.54 ranks first among the ten participating systems (Table 2), 4.53 above the runner-up. Figure 3 (right) gives a qualitative view of a 16-camera scene. On the shown frame, the predictions recover 10 of 10 PalletTrucks and 7 of 9 Forklifts across the roughly 100 m scene extent; most remaining misses occur in the dense Person class. Figure 4 projects the same world-frame predictions and ground truth into two camera views. Their agreement provides a direct visual check that the inverse transform is consistent with the site calibration.

Table 1: Scored configurations on the final full hidden test set. The upper block is the submitted progression, the lower block lists explored or discarded variants. Every row changes one variable relative to its control (Sec. 4.1); bold denotes the submitted best. Metrics are the server-displayed values and are aggregated independently.

configuration	DetA	AssA	LocA	HOTA
GaugeAlign base (canonicalization + protocol)	54.94	49.15	79.46	56.00
+ temporal adaptation λ=0.9	55.34	49.38	79.49	56.37
+ temporal adaptation λ=0.95 (submitted)	55.64	49.39	79.56	**56.54**
explored / discarded				
per-class row-select ensemble	55.07	49.46	79.56	56.33
cross-view consensus rule, one scene	55.31	49.37	79.48	56.35
cross-view consensus rule, all synthetic scenes	55.17	49.13	79.47	56.13
gap interpolation + NMS (non-causal, discarded)	55.21	49.44	79.58	56.38
size-prior variant (flawed)	51.20	49.90	77.50	53.78
monocular 2D baseline (offline)	4.86	16.01	46.66	7.98

Table 2: Comparison with participating systems on the final full hidden test set, top five of ten [29]. Team names and scores come from the organizer leaderboard; method descriptions were unavailable. Bold denotes the best value in each metric column. Our margin over the runner-up is DetA +10.3 and AssA −7.1.

rank	team	DetA	AssA	LocA	HOTA
1	EVA (ours)	**55.64**	49.39	**79.56**	**56.54**
2	SKKU-AL-T1	45.31	**56.50**	76.24	52.01
3	Playbox	40.26	31.10	75.18	38.01
4	QDTers	29.47	33.11	17.68	34.18
5	TU-YMLab	26.52	30.62	69.70	25.88

The margin is a detection margin. Table 2 decomposes the margin: DetA 55.64 against the runner-up's 45.31, while our AssA of 49.39 is 7.1 points lower (Fig. 3, left). This decomposition is consistent with a detection-focused intervention but does not show that canonicalization caused the leaderboard margin. We did not submit an uncanonicalized system to the hidden test, and checkpoint strength may also contribute. We therefore restrict causal attribution to Sec. 4.3. Association is analyzed in Sec. 4.5.

The temporal prior is a small, split-dependent gain. λ=0.95 scores 56.54 against 56.37 for λ=0.9 on the full set, and the order reversed on the 50% development split (58.73 against 58.92). This $+0.17/-0.19$ reversal is comparable to split variation; we report the decay as competition tuning, not as a robust method gain. Other variants traded improvements in one metric for losses in another: the per-class ensemble raised AssA but lowered DetA, while gap interpolation scored 56.38 but used future frames and was discarded.

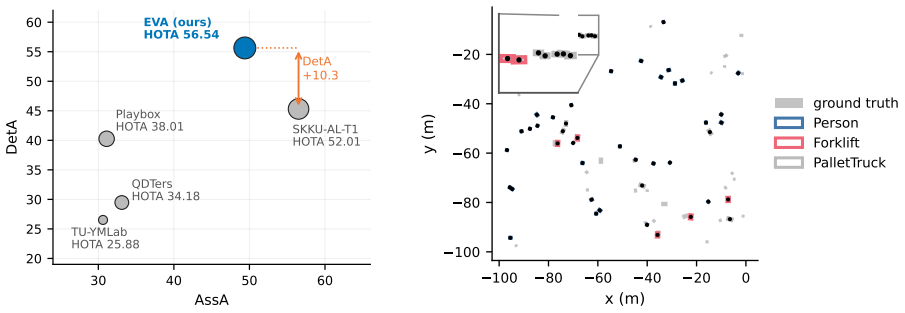


Fig. 3: Left: top five systems as DetA against AssA, with marker area proportional to HOTA. Our entry leads on detection and trails the runner-up on association. **Right:** qualitative result with canonicalization active (Warehouse 020, 16 cameras, busiest frame, bird's-eye view). Gray boxes are ground truth, colored outlines are predictions, and the inset magnifies the densest region.

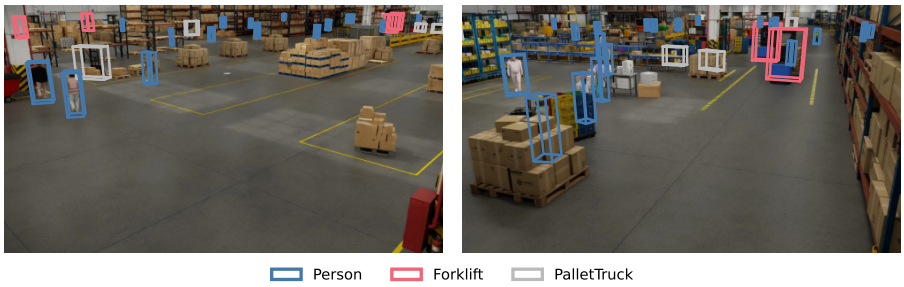


Fig. 4: Qualitative results in image space (Warehouse 020, frame 528, two of 16 views). World-frame 3D predictions of the canonicalized detector are projected with the site calibration; solid colored boxes are predictions and white boxes are ground truth. No image-space supervision or per-camera tuning is involved.

Contrast with a monocular baseline. Our monocular baseline trains a per-camera 2D detector on synthetic data and lifts its outputs to 3D with a ground-plane homography. Its DetA fell from 21.6 on synthetic validation to 4.86 on the hidden test set, whereas the multi-view system showed no comparable decline. Because the comparison also differs in architecture, pretraining data, and capacity, we report it only as an observation.

4.3 Controlled studies: the alignment is causal

A strong pretrained detector at the top of a benchmark leaves open whether the alignment mattered. We therefore isolate it on validation scenes (150 frames each, all model state fixed). These studies establish a within-model effect; they do not explain the hidden-test margin over other systems.

Table 3: Gauge stress curve. Detection-weighted aggregate DetA on three validation scenes as the world frame is displaced by δ meters (δ=0: canonical gauge, δ=$|\bar{C}_{xy}|$: absolute coordinates). Bold marks the best value in each summary row. Every entry is one inference run.

scene (cameras)	δ=0	δ=10	δ=25	δ=50	δ=$\lvert\bar{C}_{xy}\rvert$
W020 (16)	39.56	39.61	39.06	36.57	21.90
W021 (16)	26.77	26.74	26.01	23.73	14.86
W022 (4)	42.16	41.27	40.54	34.16	0.00
mean DetA	**36.16**	35.87	35.20	31.49	12.25
mean HOTA	**44.93**	44.52	44.16	40.66	19.09

Table 4: Unlock factorial on W022 (150 frames, detection-weighted class aggregate). "abs." denotes the uncanonicalized frame; bold denotes the best row. BGR is the submitted TAO input convention. Each cell is one inference run.

canon.	color	remap	dets/frame	DetA	HOTA
✓	BGR	✓	5.7	**41.99**	**47.20**
✓	BGR	–	5.6	39.17	46.16
✓	RGB	✓	5.6	41.23	46.49
✓	RGB	–	5.6	38.25	45.39
abs.	any	any	0.0	0.00	0.00

Gauge stress curve. We sweep a scalar distance δ from the canonical gauge toward the absolute frame along $-\hat{\bar{C}}_{xy}$, so that δ=0 denotes the canonical gauge and δ=$|\bar{C}_{xy}|$ denotes the absolute frame of naive transfer. Images, weights, anchors, thresholds, and decoded frames are identical at every δ; ground truth is scored in the absolute frame.

Table 3 shows that canonicalization is decisive on W022 and beneficial on average: at absolute coordinates mean DetA falls from 36.16 to 12.25, and the four-camera scene produces no valid detections. The mean moves by less than one point out to δ=25 m, and degradation becomes steep only past 50 m (Fig. 5, left). This demonstrates local tolerance around $\bar{C}_{xy}$, not equivalence of untested center choices. Two caveats apply: the sweep samples one translation direction, and the smaller degradation in the two 16-camera scenes is only a correlation across three scenes with different content, not controlled evidence for an effect of camera count.

Unlock factorial. Table 4 crosses canonicalization, the class remap, and the BGR-to-RGB conversion, an implementation-level factor, on the four-camera scene. All four cells without canonicalization emit zero detections. With canonicalization enabled, removing the class remap costs 2.83 DetA, and converting BGR to RGB costs 0.76, roughly additive in combination (−3.74). The gauge term therefore outweighs the largest secondary single-factor drop by $41.99/2.83 \approx 14.8\times$ (Fig. 5, right). The remap has the larger effect because it assigns rare classes to the wrong

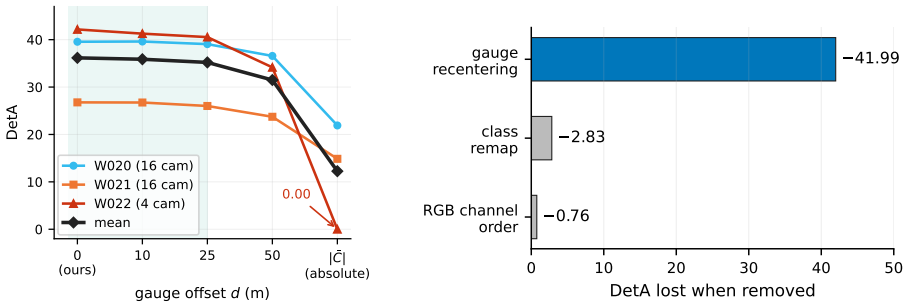


Fig. 5: Left: detection accuracy (DetA) at the five settings in Table 3. Markers are equally spaced categories, not metric distances; the last is 86 m for W020/W021 and 104 m for W022. Plot labels d and $|\bar{C}|$ denote δ and $|\bar{C}_{xy}|$. **Right:** W022 single-factor drops. The legacy "RGB channel order" bar is the 0.76-point loss from changing submitted BGR to RGB. The 14.8× ratio compares the W022 collapse with its class-remap drop, not a cross-scene effect.

evaluation categories, whereas Person maps to itself. The collapse of the absolute-frame cells prevents estimation of protocol effects without canonicalization, and the factorial was not repeated on W020/W021. Table 3 provides the separate three-scene evidence for the gauge effect. The W022 δ=0 entries in Tables 3 and 4 are separate GPU runs: their outputs contain 850 and 849 rows, giving 42.16 and 41.99 DetA. We retain both raw results; the 0.17-point difference is an observed run-to-run variation, and no multi-seed uncertainty estimate is available.

4.4 Inference-time alignment versus fine-tuning

The alternative to inference-time alignment is fine-tuning on the target-year synthetic data. With the released configuration, fine-tuning causes severe over-prediction: within tens of steps the number of detections per frame rises from about 45 to 200, and Person DetA falls from 48.9 to 1.2 in a full run. A major contributor is the learning rate. The released rate (10^{-4}) was calibrated for a global batch of 128; reducing it to 6×10^{-6} at our batch of 4 prevents the early increase in detection count in our controlled comparison (Fig. 6, left).

A second factor is plausible but not isolated. The stock set-prediction loss discards unmatched background queries, removing the gradient that suppresses false positives. Restoring this term reduced first-epoch over-prediction by 7.3× and kept Person accuracy near its baseline during a full fine-tuning run. However, this configuration also differs in learning rate, data mix, and regularization, so the contribution of the loss term is not isolated. Even at the reduced rate, deeper fine-tuning eroded strong classes faster than it improved rare ones, so no fine-tuned weights enter the final system. Fine-tuning also needs labeled target-domain data that the deployment setting does not provide.

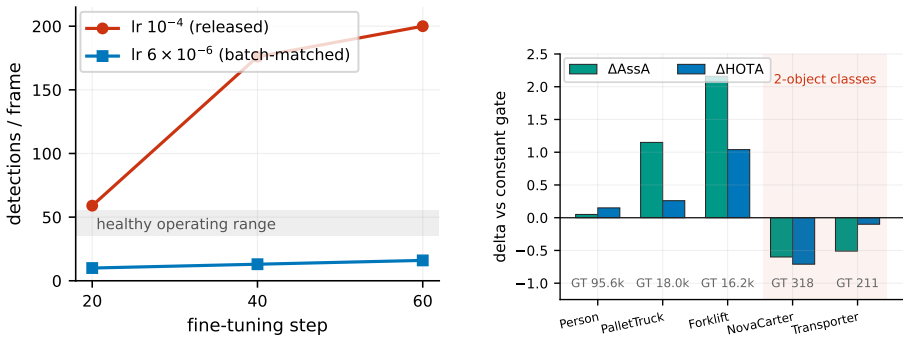


Fig. 6: Left: fine-tuning sensitivity to learning rate, with all else fixed. The released rate (10^{-4}, calibrated at batch 128) reaches 200 detections per frame; the reduced rate (6×10^{-6}) stays near the baseline band. The original label "batch-matched" denotes this heuristic reduction, not exact linear scaling from batch 128 to 4. **Right:** learned survival gate against the constant-gate baseline on the validation scenes, classes ordered by ground-truth detection count. The gate improves the three well-populated classes and degrades the two classes represented by only two objects each (shaded), where one identity flip dominates.

4.5 Association analysis: the boundary of the framework

GaugeAlign repairs detection, and Table 2 shows association is where we trail. Three probes explain why this axis resisted label-free improvement.

A hand-crafted consensus rule reduces AssA. We first built an online rule that reweights the instance memory according to cross-view support, allowing instances observed by more cameras to persist longer. Cross-view support is predictive: measured against an independent RT-DETR [38] 2D detector, it separates true from false boxes with AUC 0.873 (0.892 in the low-confidence band). As a survival rule, however, it was net-negative on the hidden test set: 56.35 on one scene and 56.13 on all three synthetic scenes against the 56.54 baseline, with aggregate AssA falling from 49.39 to 49.13. A likely reason is that a fixed threshold over-merges when many views vote, injecting identity switches into the memory.

A learned survival gate yields inconsistent class-wise gains. We insert a small MLP into the instance bank. It predicts a survival gate $g \in (0.81, 0.99)$ for each instance, turning the constant decay of Sec. 3.5 into a state-dependent one. Its input is a 59-dimensional feature vector per instance, covering the temporal state of the bank slot, the 3D geometry of the current box, cross-view consensus scalars against the independent 2D detector, and a frozen random projection of the identity embedding. In the released implementation, age, time gap, and speed are zero-filled, leaving 56 informative dimensions. We train only the MLP, using survival labels obtained by matching emitted bank instances to ground-truth tracks on eight training scenes. A label is positive if the matched

track persists for the next five sampled frames. To limit regressions, the head is initialized to reproduce the constant gate, and a checkpoint is retained only if it outperforms that baseline on scene-disjoint held-out data. It does so, raising survival-prediction AUC from 0.500 to 0.9965. This target is not the downstream HOTA objective. Because current/previous confidence and bank state help define which emitted slot survives, the AUC may exploit a shortcut; we did not run a leakage-controlled feature ablation and do not treat the AUC as evidence of tracking improvement.

On frames 0–899, the gate improves the well-populated classes: Forklift gains 2.16 AssA and 1.04 HOTA, PalletTruck 1.15 AssA, and Person 0.05 AssA. Performance declines for the two classes represented by only two objects each (NovaCarter −0.71, Transporter −0.10 HOTA), where one identity flip dominates (Fig. 6, right). The gate therefore fails our per-class criterion and was not submitted. This result covers one tenth of the sequence, is not confirmed on the hidden test, and lacks a leakage-controlled feature ablation.

Low-confidence predictions do not support recall recovery. Non-causal gap interpolation was discarded (56.38 versus 56.54), and online confidence hysteresis reduced validation DetA at every setting (−0.21 to −0.90). Scores from 0.05 to 0.2 yield roughly 160 false positives per true positive; association likely requires a trained appearance re-identification head.

5 Limitations

GaugeAlign requires accurate target-site calibration; noise and uncalibrated deployment are untested. Our one-direction, three-scene study omits alternative centers, vertical shifts, rotation, tilt, axis, and scale changes. Evidence is limited to one detector and benchmark, with one run per setting. Fine-tuning remains confounded by differences in the training recipe; the survival probe lacks a leakage-controlled ablation and fails the per-class criterion.

Reproducibility. We use the public TAO Sparse4D ResNet-50 `deployable_v2.2` checkpoint [23], challenge data, and no fine-tuning.

6 Conclusion

On Sparse4D, learned anchors tie inference to the training frame. GaugeAlign restores horizontal-translation equivariance with a closed-form, label-free transform. Our system ranked first on the hidden test; controlled validation isolates canonicalization. Evidence covers one detector and horizontal translations only; association remains unresolved.

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